

SIDDAGANGA INSTITUTE OF TECHNOLOGY-TUMAKURU 572103**Department of Information Science and Engineering****Mini Project Synopsis for the Academic Year 2025-26**

Batch No.	:	A7			
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Guide details	:	Dr Rudramurthy M. S Associate Professor of Information Science and Engineering Siddaganga Intitute of Technology, Tumakuru.			
Title of the project	:	AgriSense+: A Smart Environmental Monitoring System for Poultry and Indoor Farming			
Problem Statement	:	Traditional poultry and indoor farming systems lack real-time monitoring of key environmental parameters like temperature, humidity, gases, and light, leading to reduced productivity and health risks. A smart system using BME690 and other sensors is needed to maintain optimal conditions and trigger alerts.			
Abstract	:	Modern poultry and indoor farming require precise environmental monitoring to ensure animal welfare and crop productivity. This project introduces AgriSense+, a smart monitoring system utilizing the BME690 sensor (for temperature, humidity, and air quality), MQ-135/MQ-137 (for ammonia and gases), light sensors, and soil moisture sensors. Designed for poultry houses and indoor farming setups, the system monitors conditions in real time and generates alerts when parameters deviate from optimal ranges. The solution supports sustainable and efficient farming through data-driven control and adaptability.			
Objectives	:	To develop a smart environmental monitoring system for poultry and indoor farming using BME690 (for temperature, humidity, and air quality), MQ-135/MQ-137 (for ammonia and gas detection), light sensors, and soil moisture			
Applications	:	* Poultry farm management and health optimization * Indoor/vertical farming environment control * Automated greenhouse climate regulation * Early detection of hazardous gases (ammonia, CO ₂) * Environmental data logging for agricultural research			
Signature of students	:				
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Name and Signature of Guide with date	:				
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